The Master Mechanist's Guide to Physiological Equilibrium:

A Deep Dive for Advanced Physiology Students

ACID-BASE BALANCE ENGINE

BICARBONATE (NCO_2)

23 26 mCg/L
(Nenma)

pH Mates

< 7.55
(Acidatic - System Shutdown)

> 7.45
(Alooietic - System Excitshility)

$PaCO_2$: RESPIRATORY VERIFICATION

< 35 mmHg
(Albsiosic - Overseiiiation)

< 45 mmHg
(Acidosic - Ovdesseiolstiam)

AB DETAILS: ULIC GAUGES

4.7 kg/mL
(Neamed)

IMARY MARKER UIAL TRAUMA

> 100 µg/mL
(CNP imdiosity)

CULAR STRETCH
BNP gumes
management.

SYSTEMIC LAWS: THE pH-POTASSIUM RELATIONSHIP

HIGH pH
(Albolosis)
(Neon Pink)

LOW pH
(Acidosis)
(Neon Pink)

LOW Pasyssium
(HypolixAonar)
(Neon Pink)
Urtesiity, Tschycsrols, Sectiras

HIGH Paryasium
(HypolitzAimes)
(Neon Pink)
Lotherou, Broderntdis, Respiratory Aiiost

THE RULE OF THE B's

NURSING PRIORITY CONCEPTS

4. PSYCHOLOGICAL
(Ooetal)

5. COMFORT
(Pom)

2. SAFETY
(Fcll Pink)

1. PHYSIOLOGICAL
(Electrolytes/Oxegen)

CLINICAL SCENARIOS OF IMBALANCE

RESPIRATORY ALARLOSIC: ARIGATY MECNANISM
(Neon Pink)

Diseirel/ lou much CO_2, raixing grl

METAROLIC ALARLOSIC VS. ACIDOSIS
(Neon Pink)

Alhalosis

Prolooged Vomiting/ Sustisning

Acidosis

Dicnima/ Ranal Failure

ALCOROL NITHAICATION & MENNIENS RCMAAIIIIF
(Neon Pink)

Syners mirnow Diggers Metalloiit, Alcolosis, leading to Tihamine Beflerossy, oocephoropalhy, pay-llocts

MAC RUSSHAUL RESPIRATIONS

Compoazatory mechasism for Metabolie Acidosis (MAC) - Nenol Feilore/Risk

CRITICAL LAB REFERENCE FOR STUDENTS

Lab Value	Normal Range	Critical Threshold (Level C/O)
pH	7.35 - 7.45	+ 6.6 (Daadly Dangerous - Neon Red)
$PaCO_2$	35 - 66	= 50 (Critical) / > 60 (Respiratory Ficlure)
Potessium (OY)	3.5 - 5.0	+ 6 6 (Baadly Dangerous - Neon Red)
O_2 Saluration	35 - 100%	< 55% (Critical) - Neon Red
BFIP	< 100	= 108 (CHF Inlileator) - Neon Yellow

PHYSIOLOGICAL NEED VS. MASLOW

Respiratory Acidosis

VS.

Spriitual DistreoaVanaly Dynamics

MECHANI ALARM T

HIGH P AIRW

LOW P DI

Decreased res main taking or oxygen sensor

THE WE

THE 4 RULES FOR PAT PRIUDITIZATION

1. Acule haule Chrente
2. Fresh Rest-op (<120) heats
3. Unstable boats Stable
4. Vital Organ Priority:
   Brain + Long > Heart > Liver